



Covariance matrix reverse-engineering for extreme scenario stress testing

Marie Curie Presentation

July, 30th

Ghislain Gauthier – MSCI Inc. (Industry Host)

**(FP7) MC ITN-RISK-2009-25:
“Long Horizon Portfolio Volatility”**

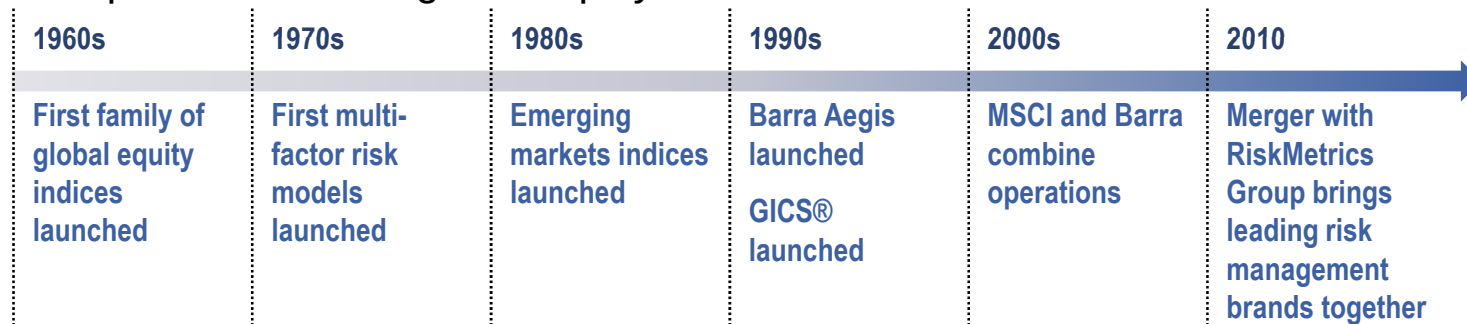
Outline

- MSCI and the Marie Curie research project
- Correlation and extreme events
- Manipulation correlation in the Barra Global Equity Model – GEM2 – framework
- Reverse stress-testing: covariance matrix reverse-engineering on historical scenarios.

MSCI – A Clear View of Risk & Return

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- **Large and experienced** team of index, portfolio analytics and governance specialists
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- Supporting **thousands of clients worldwide**, including asset managers, banks, hedge funds and pension funds

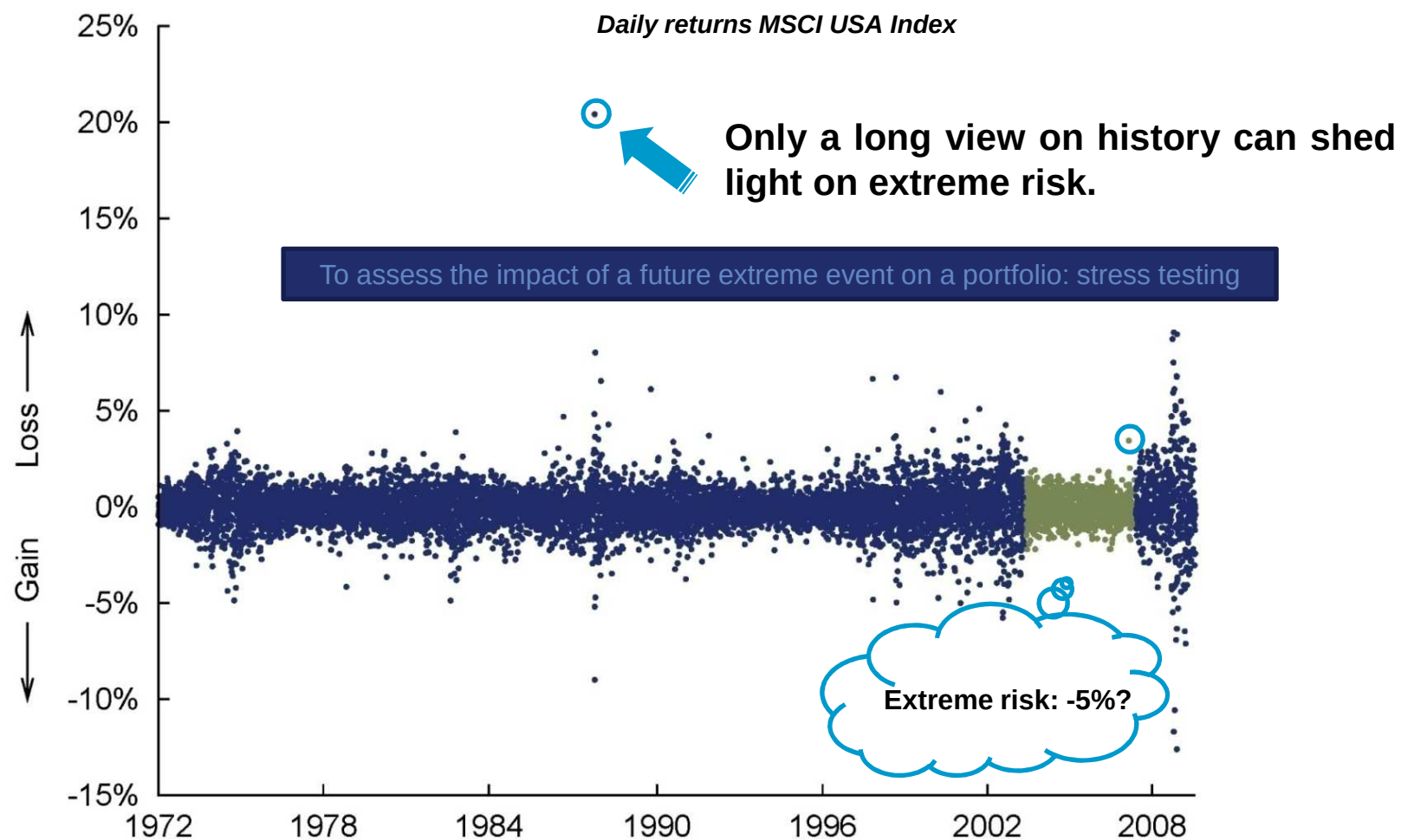
Marie Curie ITN – “Long Horizon Portfolio Volatility”

- Start: Nov 2009
- 12-month project, including secondment to MBS (Sept-Oct 2010)
- Working within the EMEA Applied Research team, based in Geneva
- Collaboration with:
 - Global Applied Research team (London, New York)
 - Risk Analytics research team (Berkeley, Geneva, Budapest)
 - Other Marie Curie network partners
 - Very recently: RiskMetrics research

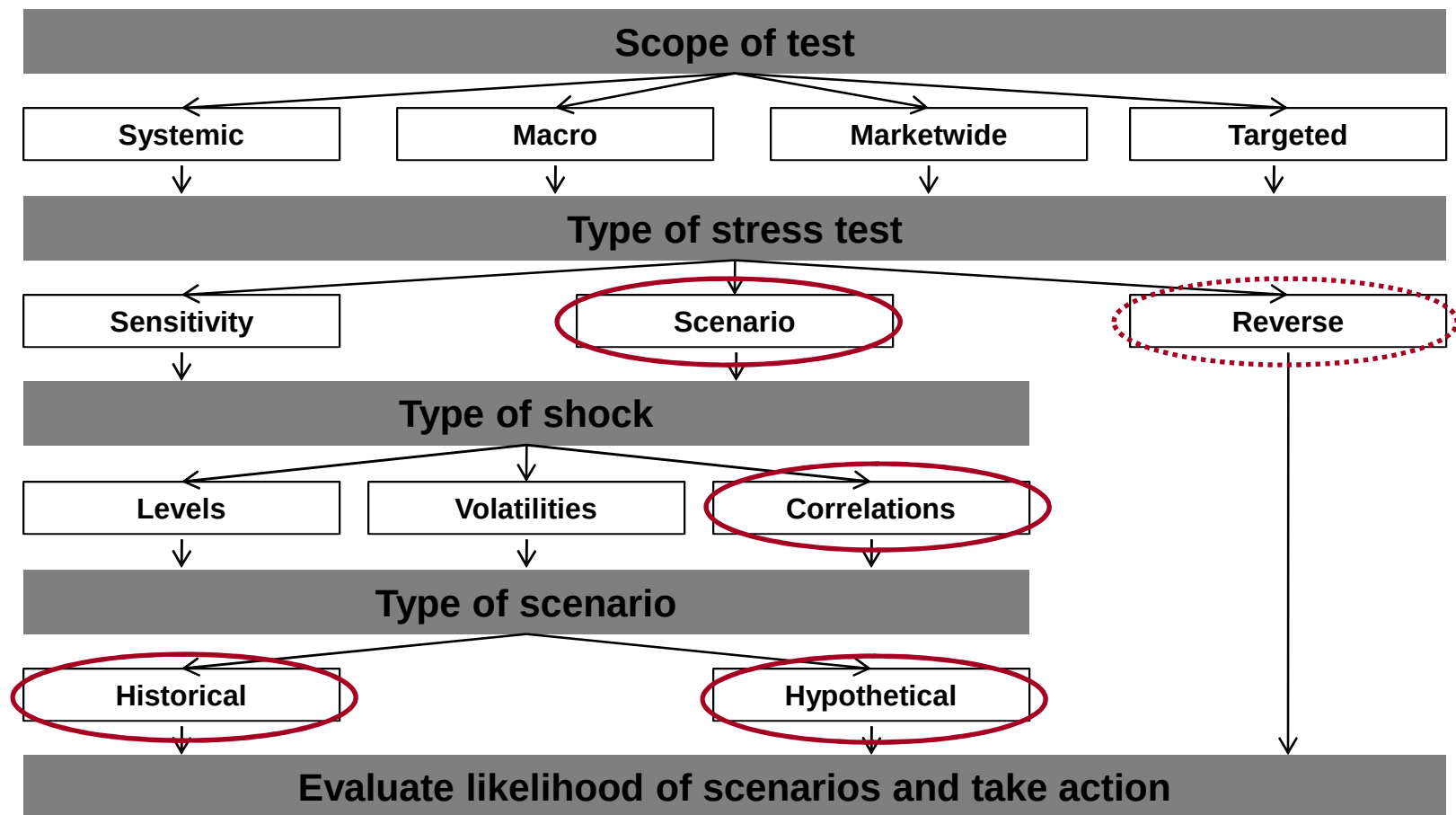
Objectives

- Assess the impact of future extreme events on multi-asset-class portfolios
- Construct a better stress-testing framework for portfolio managers, providing functionality such as user-defined strategic asset class assumptions and stress testing scenarios.
- Contribute to new stress test standards for the asset management industry
- Implement the new MSCI latent factor methodology to stress correlations, and research its application to historical extreme scenarios

Risk is Multi-faceted



What is Stress-testing?

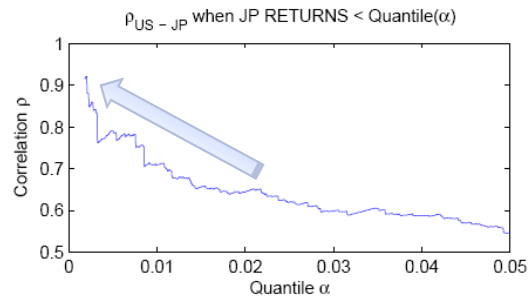
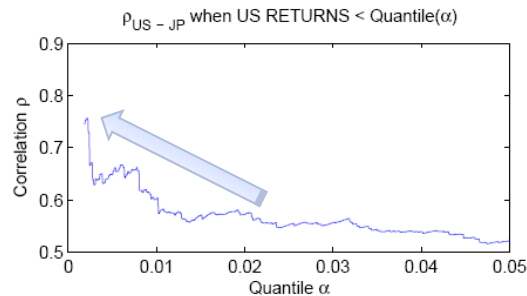


Incorporating Extreme Correlations

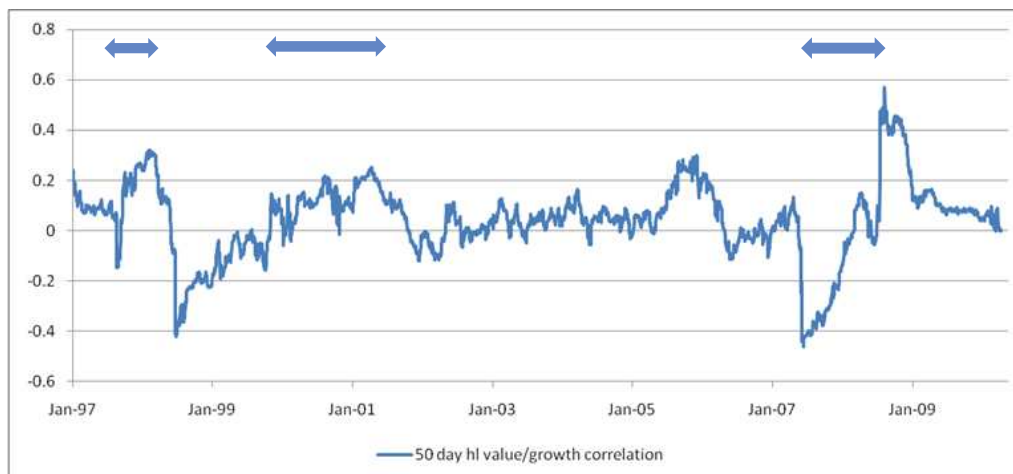
- The ability to change correlations manually addresses many of the issues:
 - The risk manager can easily incorporate hypothetical extreme changes where correlations “go to 1”
 - The risk manager doesn’t have to rely solely on historically observed correlation matrices (and avoids some of the estimation issues)
 - The risk manager can incorporate views or forecasts of correlations for which he has information

Correlations and extreme events

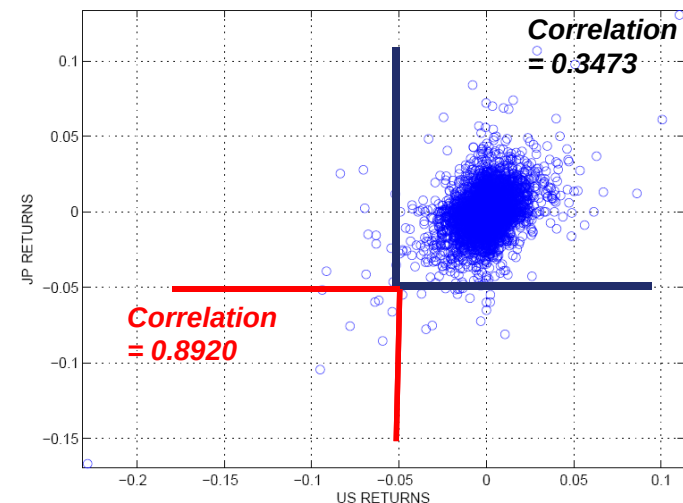
Correlations and Extreme Events



During extreme events, correlations across assets increase dramatically (as the extreme losses tend to occur often jointly). Thus, a diversified portfolio can become a highly correlated asset portfolio and lead to extreme losses.



MSCI Japan vs. MSCI US (1 day lag)
Daily returns : 1986-2009



Extreme event is defined as a daily return of -5% in both markets

Risk Multi-Factor Models

- Identify the factors that *independently* explain returns
- For each asset, assign exposures to each factor
- Use statistical techniques to...
 - Estimate the returns for each factor each month
 - Estimate the covariance between factor returns
- How good is the model?
 - We know the factor structure is well-specified when specific risk is uncorrelated

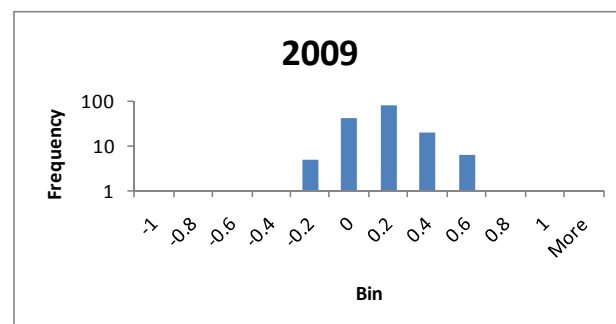
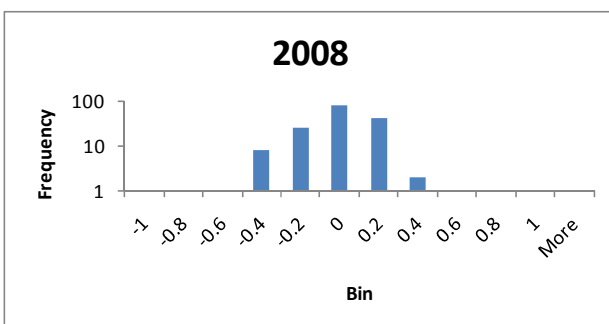
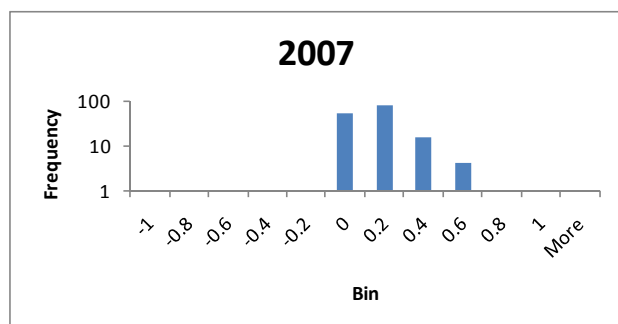
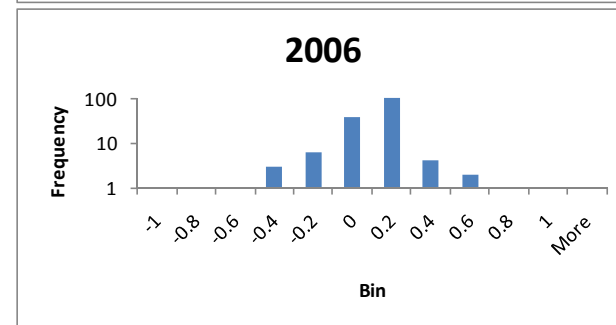
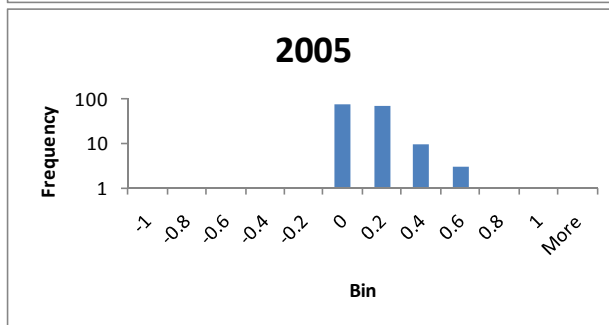
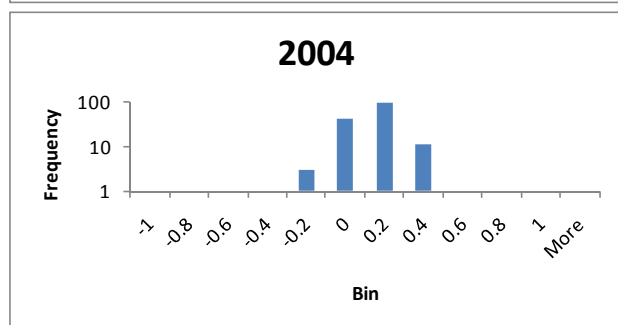
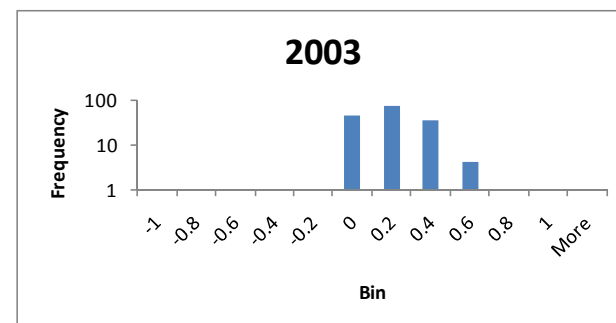
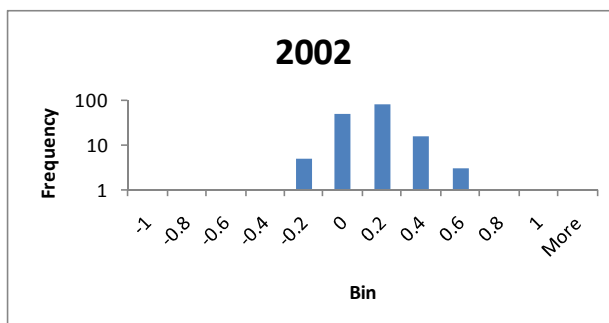
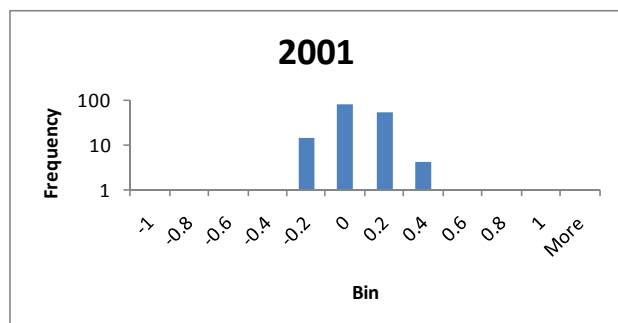
Barra Global Equity Model (GEM2)

- The equity factor set in GEM2 includes a World factor (w), countries (c), industries (i), and styles (s)

$$r_n = f_w + \sum_c X_{nc} f_c + \sum_i X_{ni} f_i + \sum_s X_{ns} f_s + u_n.$$

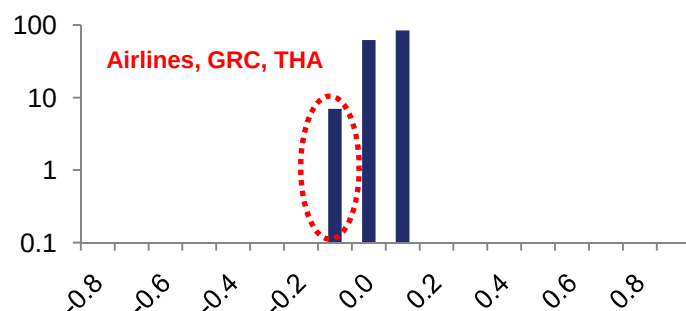
- Industry factors explain a large portion of returns:
 - Too much variation within sectors
 - Large firms are exposed to multiple industry factors
- Style (or Risk) factors reveal hidden bets on fundamental styles such as leverage, size, or momentum
- Country factors explain a large portion of global returns
 - Japan risk is different from China risk
 - Captures the majority of emerging market risk today
- Currencies are always a factor when investing in multiple markets

Distribution of GEM2 Factor Returns in Random Long Horizon Periods

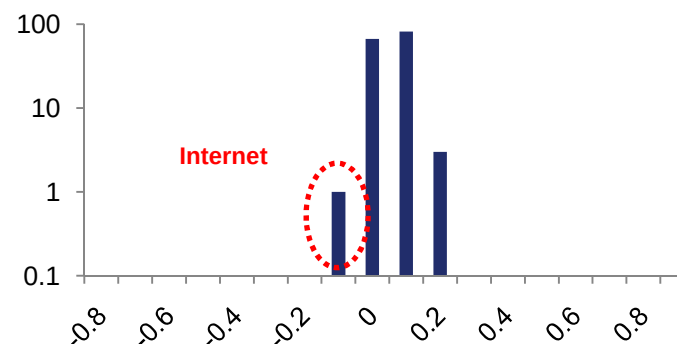


Distribution of the GEM2 factor Returns during Historical Extreme Scenarios

2001 Sept 11 (week)

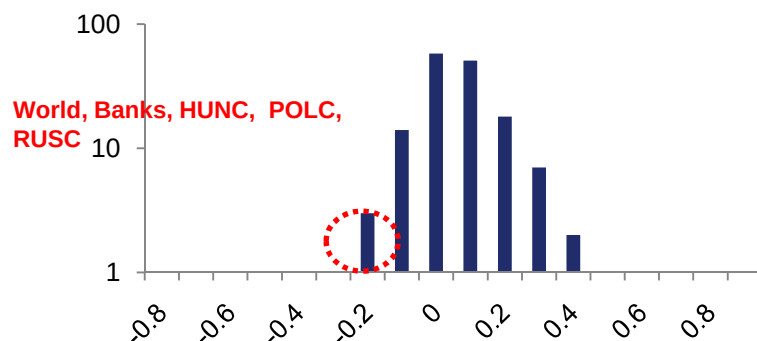


2000 Tech Bubble: Apr, 7th – Apr 14th

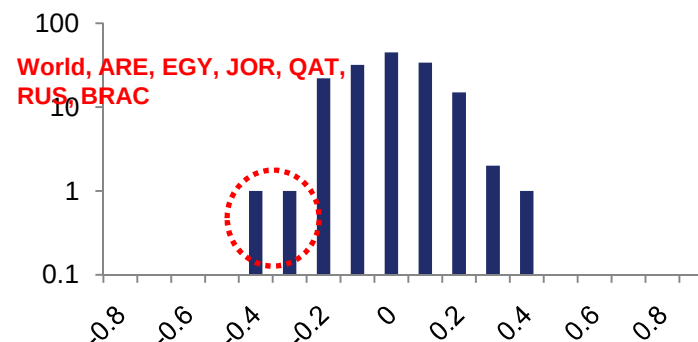


During extreme market events: a limited number of extreme negative return. However, the average of the factor returns is negative or lower than “regular” times.

2009 Jan-Mar

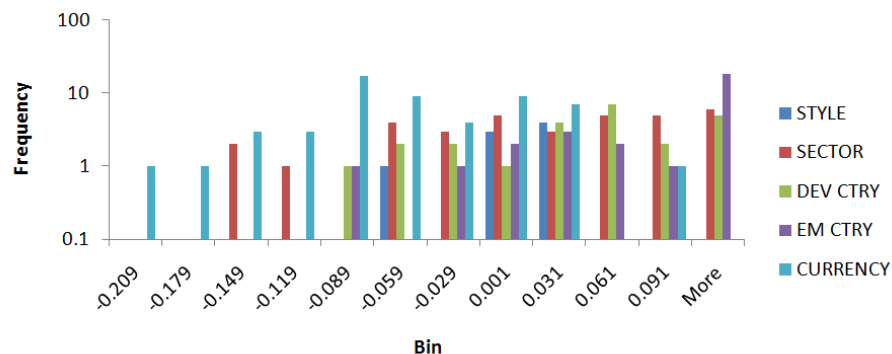


2008 Sept-Nov

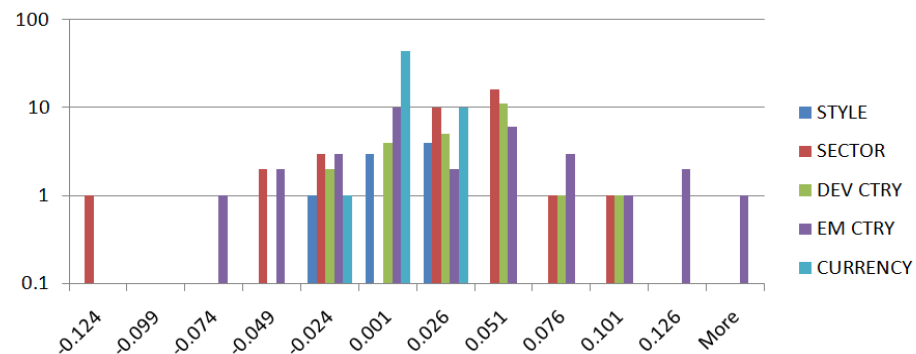


Distribution / block

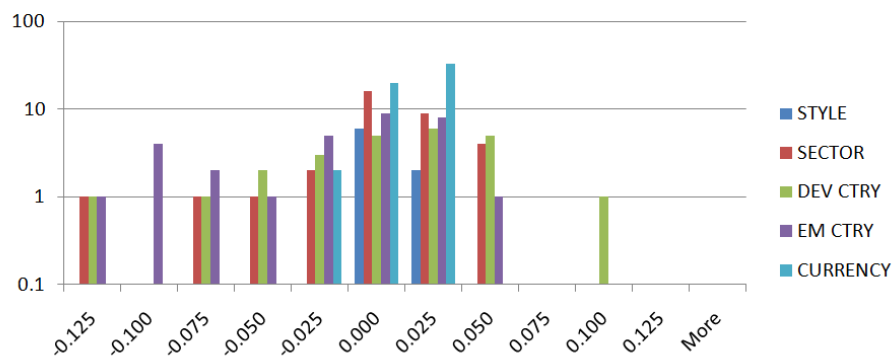
2009 Jan-Mar : Block distribution



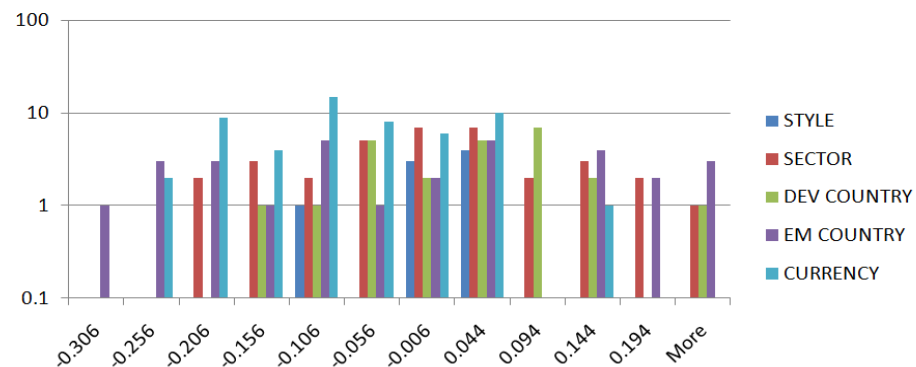
2000 Tech Bubble Block Distribution



2001 Sep 11: Block Distribution



2008 Sept-Nov Block Distribution



- In every scenario, a few factors go extremely negative
- These factors belong to different types (Style, Sector, Country, Currency) ; no clear pattern



Reverse Stress Testing

And manipulation correlations

Stress test with Historical Covariance Matrix

- Full historical stress test: specify all factor shocks, assess these impacts on the portfolio
- Correlated stress test: specify one or several factor shocks, use covariance matrix to assess the shock propagation or “contagion” to other factors, assess impact on the portfolio
- A covar matrix is calculated with a specified half-life, here we use long horizon models (half-life 6 to 48 months)
- Conditional expectation value (Multivariate normal framework):

$$E(Y | S) = E(Y) + \Sigma_{SY}^T \Sigma_{SS}^{-1} (S - E(S))$$

For our purpose, we assume

$$E(S) = 0 \text{ and } E(Y) = 0$$

n is the number of user defined shocks

m is the number of targets to be analyzed

S is a $n \times 1$ vector for user defined shocks

Y is a $m \times 1$ vector for changes in returns of the targets to be analyzed

Σ_{SS} is the $n \times n$ covariance matrix for components in S

Σ_{SY} is the $n \times m$ covariance matrix for covariance between components in S and Y

Example : Aug-Sep 2008 using Aug 2008 BIM Covar Matrix

We applied a -25% shock on each equity (est. universe), using correlated shock option in BarraOne

01/08/2008 - 30/10/2008		Analysis Date 01/08/2008				
Shock on US_EQUITY	-25%	Asset Name	Initial Market Value	test_US_EQ		Asset Id
				Final Market Value	P&L	
Portfolio	-26.06%	Total	100.00	81.01	-18.99%	
MSCI EMU Historical Return	-23.41%	MSCI Europe	33.33	25.76	-22.73%	MSEUR
MSCI JP Historical Return	-29.92%	MSCI Japan	33.33	30.07	-9.80%	MSCIJP
MSCI US Historical Return	-24.85%	MSCI USA	33.33	25.19	-24.43%	MSCIUSA

01/08/2008 - 30/10/2008		Analysis Date 01/08/2008				
Shock on JP_EQUITY	-25%	Asset Name	Initial Market Value	test_JP_EQ		Asset Id
				Final Market Value	P&L	
Portfolio	-26.06%	Total	100.00	87.78	-12.22%	
MSCI EMU Historical Return	-23.41%	MSCI Europe	33.33	31.29	-6.14%	MSEUR
MSCI JP Historical Return	-29.92%	MSCI Japan	33.33	24.78	-25.67%	MSCIJP
MSCI US Historical Return	-24.85%	MSCI USA	33.33	31.72	-4.84%	MSCIUSA

01/08/2008 - 30/10/2008		Analysis Date 01/08/2008				
Shock on EMU_EQUITY	-25%	Asset Name	Initial Market Value	test_EU_EQ		Asset Id
				Final Market Value	P&L	
Portfolio	-26.06%	Total	100.00	83.77	-16.23%	
MSCI EMU Historical Return	-23.41%	MSCI Europe	33.33	25.42	-23.73%	MSEUR
MSCI JP Historical Return	-29.92%	MSCI Japan	33.33	30.55	-8.35%	MSCIJP
MSCI US Historical Return	-24.85%	MSCI USA	33.33	27.79	-16.62%	MSCIUSA

Observation

- By shocking one of the asset and using the historical covariance matrix (right before the crisis), we can not replicate entirely the historical returns.
- Let's try using with the historical covariance matrix at the end of the period. It may incorporate the new co-movements across the assets.

Example : Aug-Sep 2008 using Nov 2008 BIM Covar Matrix

We applied a -25% shock on each equity (est. universe), using correlated shock option in BarraOne

01/08/2008 - 30/10/2008		Analysis Date 01/12/2008				
Shock on US_EQUITY	-25%	Asset Name	Initial Market Value	test_US_EQ		Asset Id
				Final Market Value	P&L	
Portfolio	-26.06%	Total	100.00	82.31	-17.69%	
MSCI EMU Historical Return	-23.41%	MSCI Europe	33.33	26.66	-20.02%	MSEUR
MSCI JP Historical Return	-29.92%	MSCI Japan	33.33	30.45	-8.64%	MSCIJP
MSCI US Historical Return	-24.85%	MSCI USA	33.33	25.20	-24.41%	MSCIUSA

01/08/2008 - 30/10/2008		Analysis Date 01/12/2008				
Shock on JP_EQUITY	-25%	Asset Name	Initial Market Value	test_JP_EQ		Asset Id
				Final Market Value	P&L	
Portfolio	-26.06%	Total	100.00	87.48	-12.52%	
MSCI EMU Historical Return	-23.41%	MSCI Europe	33.33	31.54	-5.39%	MSEUR
MSCI JP Historical Return	-29.92%	MSCI Japan	33.33	24.71	-25.87%	MSCIJP
MSCI US Historical Return	-24.85%	MSCI USA	33.33	31.23	-6.31%	MSCIUSA

01/08/2008 - 30/10/2008		Analysis Date 01/12/2008				
Shock on EMU_EQUITY	-25%	Asset Name	Initial Market Value	test_EU_EQ		Asset Id
				Final Market Value	P&L	
Portfolio	-26.06%	Total	100.00	83.35	-16.65%	
MSCI EMU Historical Return	-23.41%	MSCI Europe	33.33	25.67	-22.98%	MSEUR
MSCI JP Historical Return	-29.92%	MSCI Japan	33.33	30.76	-7.72%	MSCIJP
MSCI US Historical Return	-24.85%	MSCI USA	33.33	26.92	-19.24%	MSCIUSA

Observations

- Using the partially updated covariance matrix doesn't improve the replication of the historical returns.
- We need to shock the covariance matrix in such a way that it will reproduce the historical returns from a factor shock/a set of factor shocks.

Modelling Correlation Shocks with Latent Factors

- Consider two “seemingly” uncorrelated risk factors

$$\begin{cases} f_1 = \bar{f}_1 + S_t \cdot Z_t \\ f_2 = \bar{f}_2 + S_t \cdot Z_t \end{cases} \quad \begin{array}{l} \swarrow \text{latent factor} \\ \nwarrow \end{array} \quad \begin{array}{l} S_t = 1 \text{ with probability } 1\% \\ S_t = 0 \text{ with probability } 99\% \end{array}$$

Assume $\text{cov}(\bar{f}_1, \bar{f}_2) \approx 0$ and $\sigma_Z^2 \approx 10 \sigma_{\bar{f}_i}^2$

- Unconditional correlations between factors may be low

$$\text{corr}(f_1, f_2) = (0.01) \times (0.99) \times \frac{\sigma_Z^2}{\sigma_{\bar{f}_1} \sigma_{\bar{f}_2}} \approx 10\%$$

- But event conditional correlations can be close to one

Conditional on $S_t = 0$ $\text{corr}(f_1, f_2) = 0$

Conditional on $S_t = 1$ $\text{corr}(f_1, f_2) = \frac{1}{\sqrt{1 + \sigma_{\bar{f}_1}^2 / \sigma_Z^2} \sqrt{1 + \sigma_{\bar{f}_2}^2 / \sigma_Z^2}} \approx 90\%$

A Framework with Latent Drivers

Bender, Lee, Stefek, “Manipulating Correlations Through Latent Drivers” , Research Insight Paper, MSCI

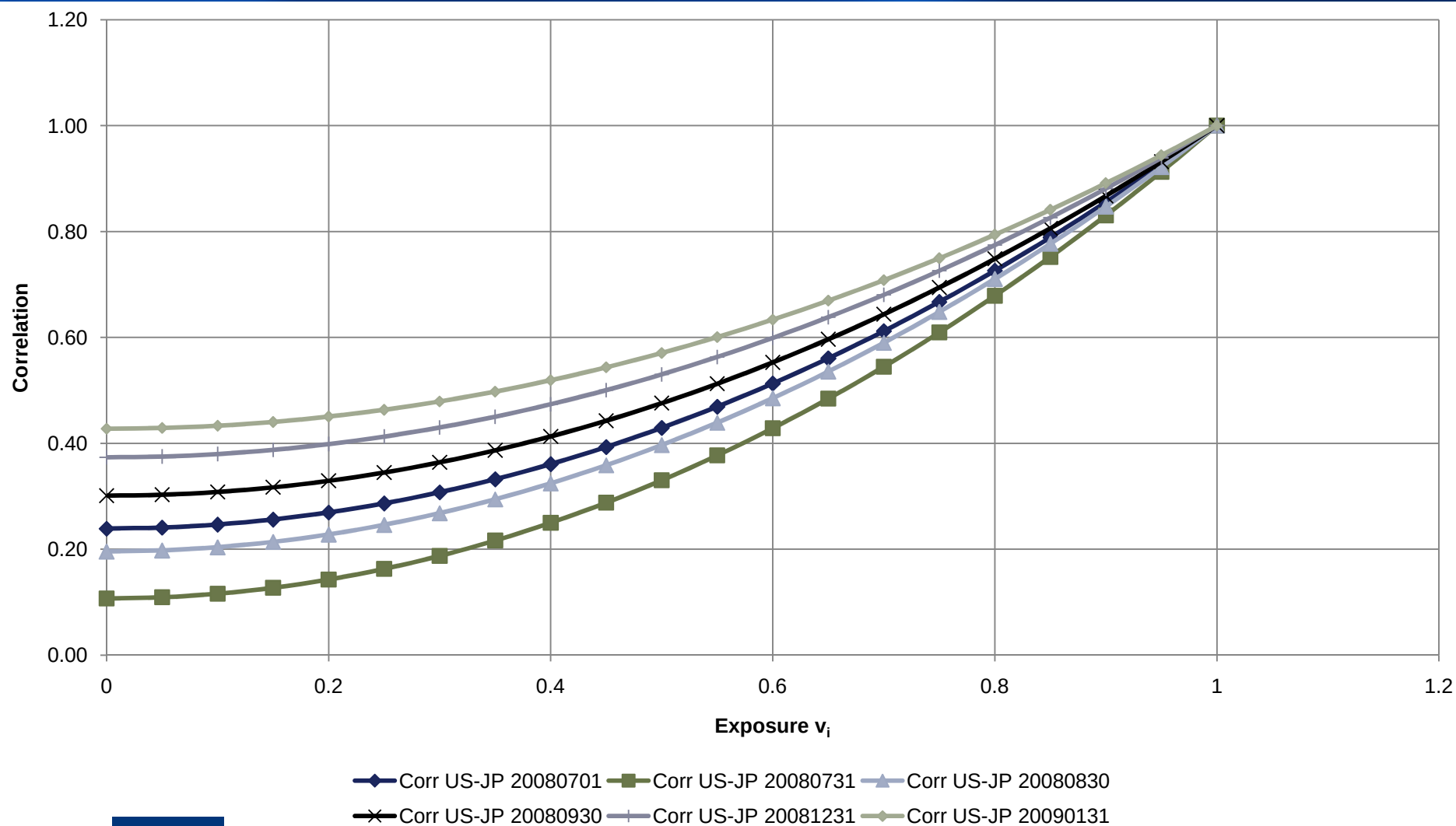
- Correlations can be manipulated through each asset's exposure to its latent driver v
- The new correlations are:

$$\rho_{ij}^{new} = v_i v_j + \sqrt{1 - v_i^2} \sqrt{1 - v_j^2} \rho_{i,j}^{orig} \quad \text{if } i \text{ and } j \text{ are in the same block}$$

$$\rho_{ij}^{new} = v_i v_j \rho_{r_1, r_2} + \sqrt{1 - v_i^2} \sqrt{1 - v_j^2} \rho_{i,j}^{orig} \quad \text{if } i \text{ and } j \text{ are in different blocks}$$

- The higher the value we assign to v_i and v_j , the more correlation will change within the block
- The higher value we assign to ρ_{r_1, r_2} , the higher correlation between blocks
- ρ_{r_1, r_2} preserves the consistency between blocks

Illustration: Manipulation of the Correlation of US-JP factors (Aug-Oct 2008 Crisis)



An Illustration: No Blocks

- When v is between 0 and 1 (for instance, 0.5)

Original correlation matrix

	US equity	Canada equity	UK equity	EMU equity*
US equity	1.00			
Canada equity	0.76	1.00		
UK equity	0.71	0.64	1.00	
EMU equity	0.73	0.67	0.79	1.00



New correlation matrix

	US equity	Canada equity	UK equity	EMU equity
US equity	1.00			
Canada equity	0.82	1.00		
UK equity	0.78	0.73	1.00	
EMU equity	0.80	0.75	0.84	1.00

$$\rho_{US,CN}^{new} = \underbrace{v_{US} v_{CN}}_{0.25} + \underbrace{\sqrt{1-v_{US}^2}}_{0.87} \underbrace{\sqrt{1-v_{CN}^2}}_{0.87} \rho_{i,j}^{orig}$$

Covariance Matrix Reverse-engineering

- Reverse-engineering a covariance matrix that is consistent with historical crises

Has to be consistent with the historical factor returns

Shock

Input

$$E(Y | S) = E(Y) + \Sigma_{SY}^T \Sigma_{SS}^{-1} (S - E(S))$$

For our purpose, we assume

$$E(S) = 0 \text{ and } E(Y) = 0$$

•2 Block version:

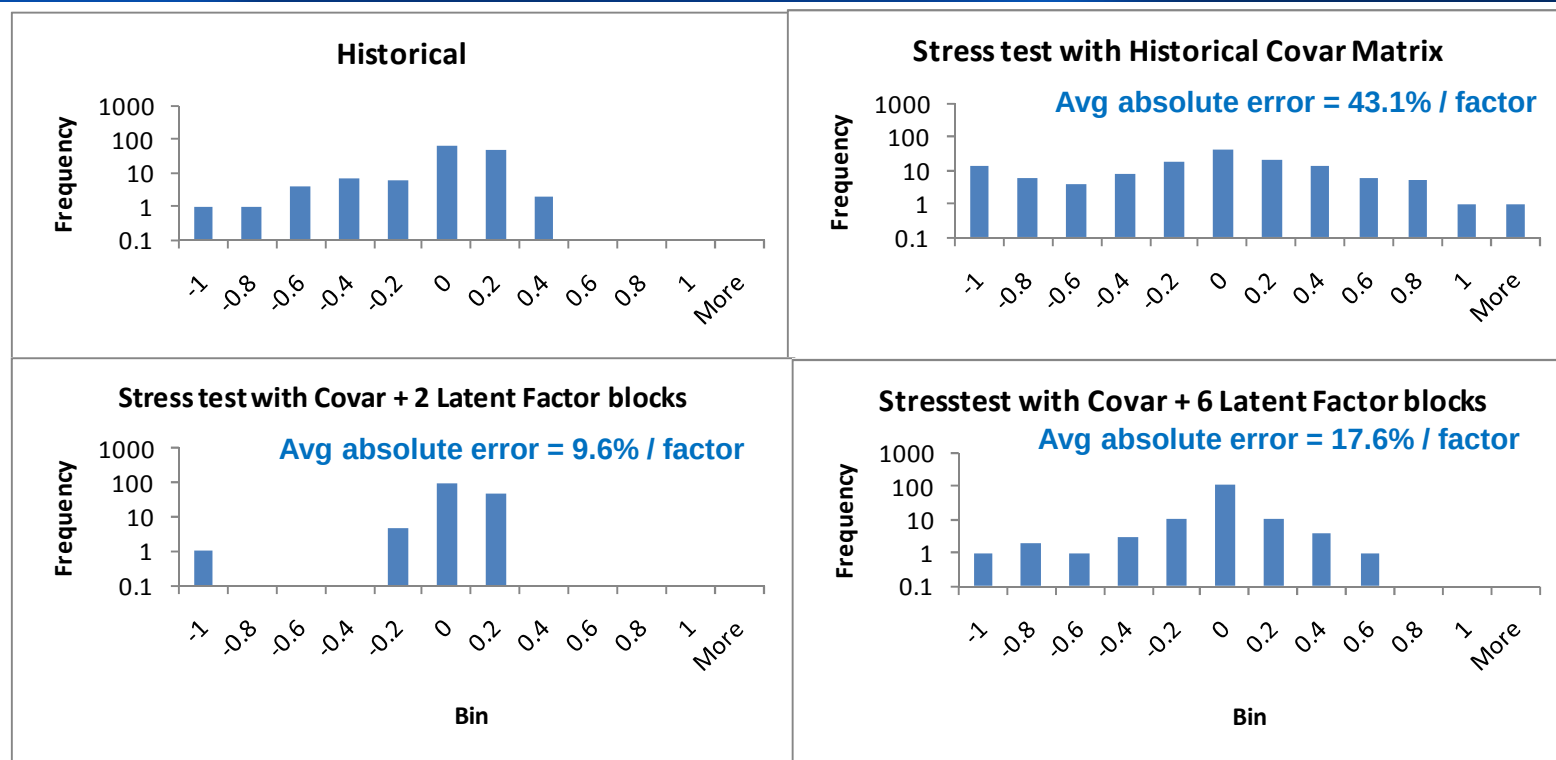
- Factors in avg – 3*sigma tail have same latent factor exposure
- Other factors have another latent factor exposure

•6 Block version:

- Factors of same type obtain same latent factor exposure (World, Styles, Sectors, DM Countries, EM Countries, Currencies)

Optimal exposures determined by looking for minimal error $\text{sum}[\text{abs}(\text{expected_return} - \text{historical_return})]$

1997-98 Asian Financial Crisis

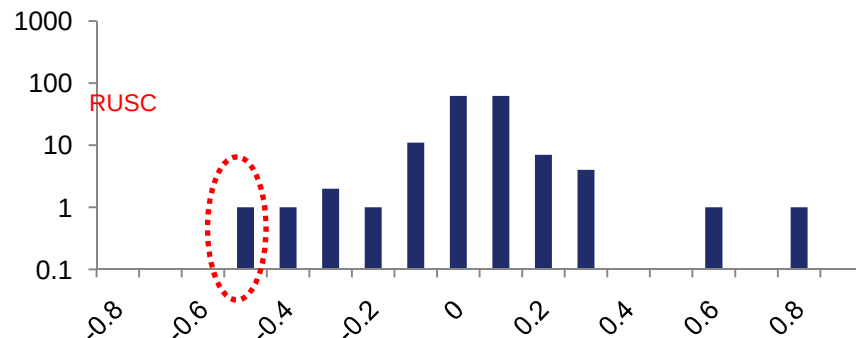


Evaluation:

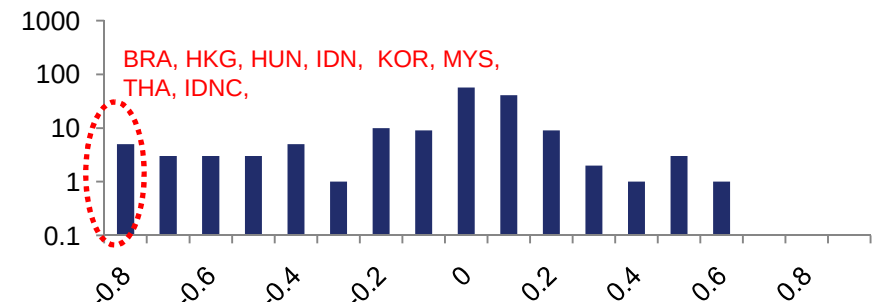
- Closer replication with latent factors than with original covar matrix, as measured by errors
- Some other factor returns in tail are not replicated yet; variation of tagged tail factors to be studied

1998 Russian Financial Crisis

Historical Returns

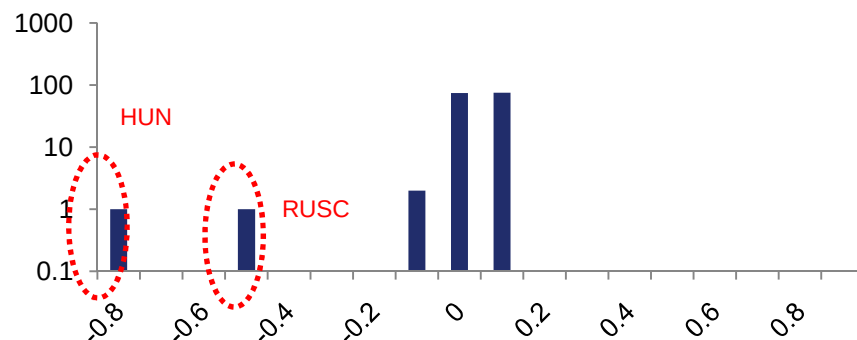


Historical covar matrix: avg abs error: 19.1% / factor

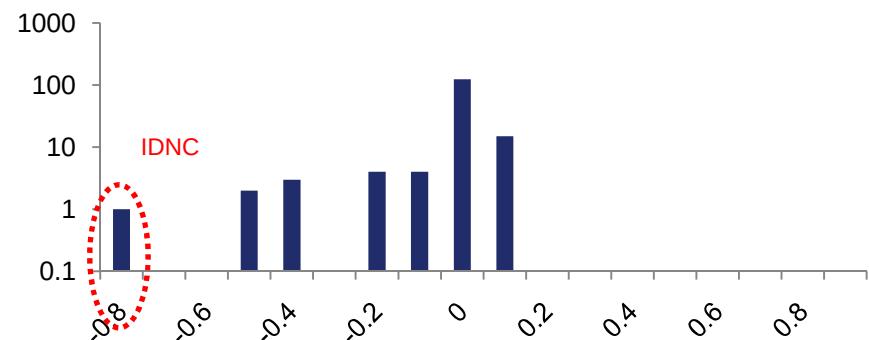


Attempt to replicate the 1998 Russian Financial Crisis historical factor returns, by finding the set of exposures that would minimize the absolute sum of the overall errors.
All the stress test lead to more extreme negative factor returns. Only the historical covar matrix replicates partially the upside of the returns, while producing the fattest negative tail.

2 block version: avg abs error: 8.8% / factor

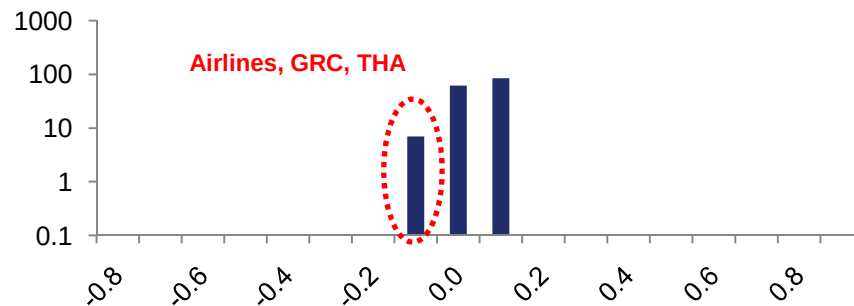


6 Block version: avg abs error: 10.4% / factor

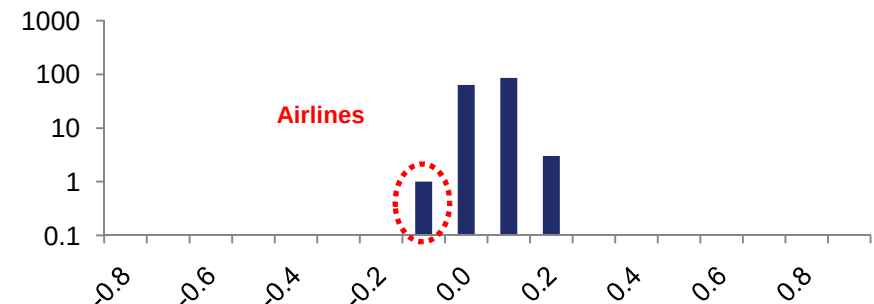


2001 Sep 11

Historical Returns



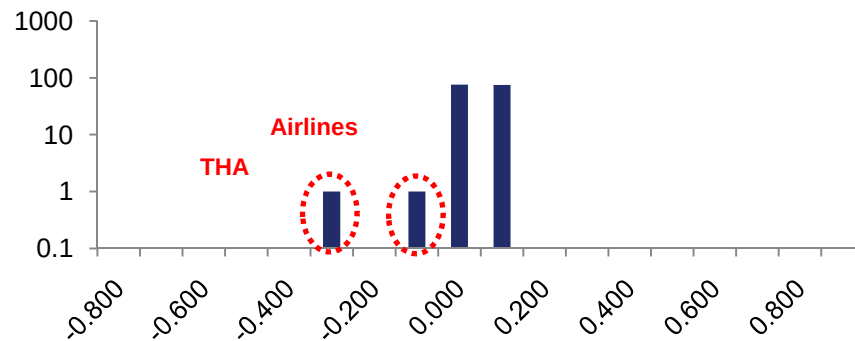
Historical covar matrix: avg abs error: 3.6% / factor



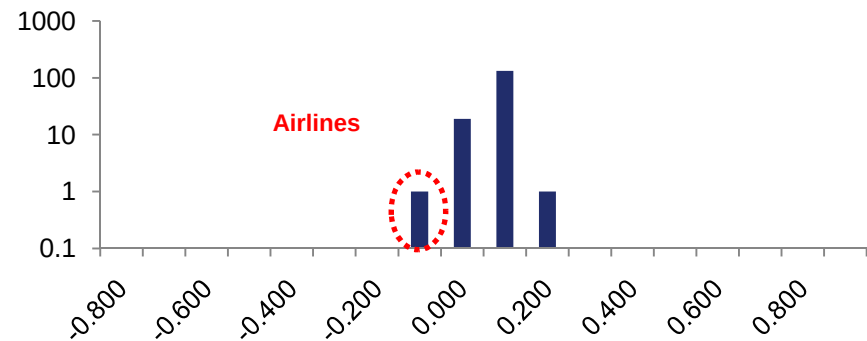
Attempt to replicate the 2001 Sept 11 (week) historical factor returns, by finding the set of exposures that would minimize the absolute sum of the overall errors.

The stress tests using the latent factor technique provide better estimations.

2 block version: avg abs error: 2.3% / factor

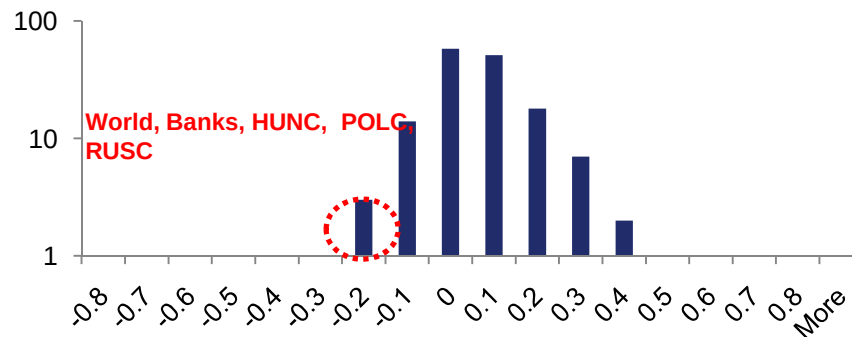


6 block version: avg absolute error: 2.4% / factor

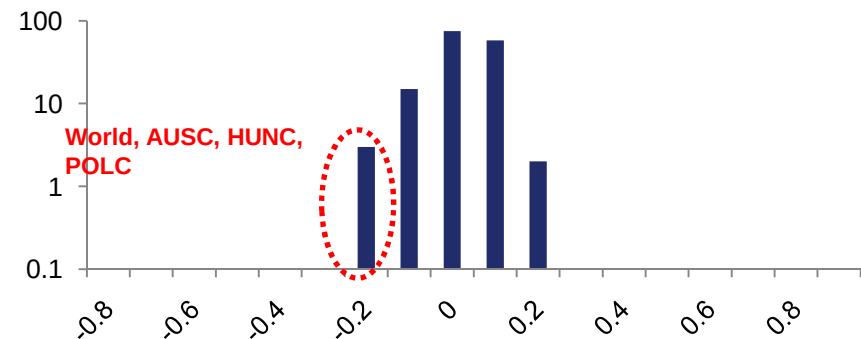


2009 Jan-Mar

Historical Returns

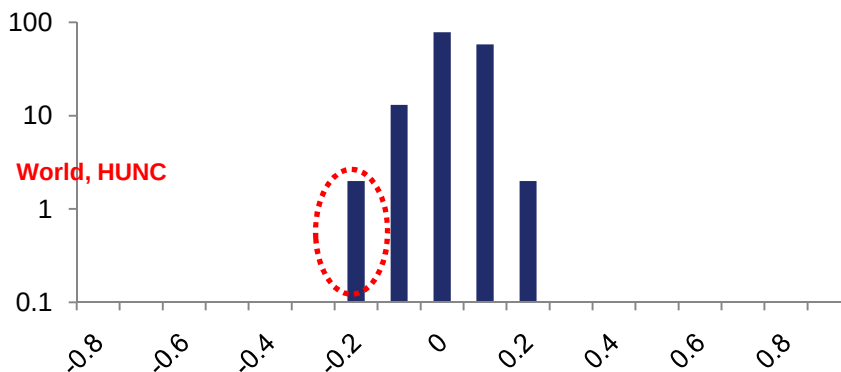


Historical covar matrix : average error: 7% / factor

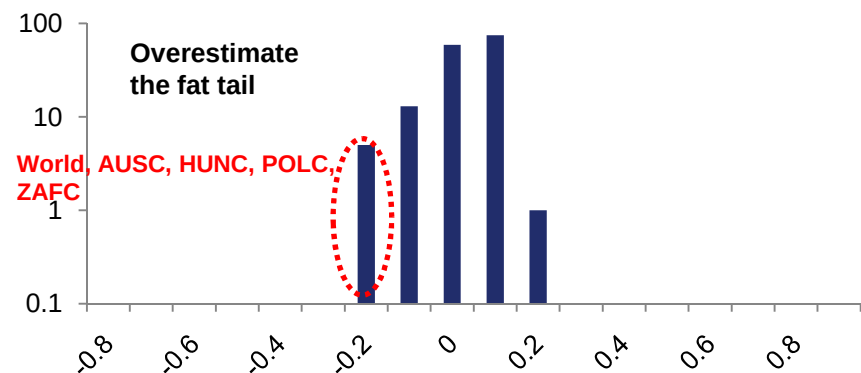


Attempt to replicate the 2009 Jan-Mar historical factor returns, by finding the set of exposures that would minimize the absolute sum of the overall errors. Can't replicate the upside of the historical distribution. All of the stress test don't replicate the upside of the returns, and lead to fatter negative tail.

2 Block version : average error : 6.9% / factor



6 Block version : average error: 6.5% / factor



Extreme Scenario Analysis Summary

- A correlated stress test using the historical GEM2 covar matrix does not reproduce the historical stressed factor returns fully
- Using a latent factor approach to stress the correlations of the covar matrix, it seems possible to create a stress test that better replicates the historical stressed factor returns
- For several scenarios, the 2 block approach (putting extremely negative factors together in one block, the other factors in a second block) produces the best result
- For some scenarios, also the 6 block approach produces an acceptable result
- For most studied scenarios the tails are not well replicated yet, including positive tails

Project Next Steps

1. Finalize content:

- Refine validating examples demonstrating the procedure updates historical covariance matrices to be consistent with historical scenarios
- 2 block version -> 3 block version:
 - As we can't replicate totally the upside of the returns, let's introduce the largest positive return in the vector of applied shocks and an exposure to it, within the covar matrix
- Create hypothetical scenario for current market concerns to motivate usefulness of this approach for practitioners

2. Disseminate results:

- Research report

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